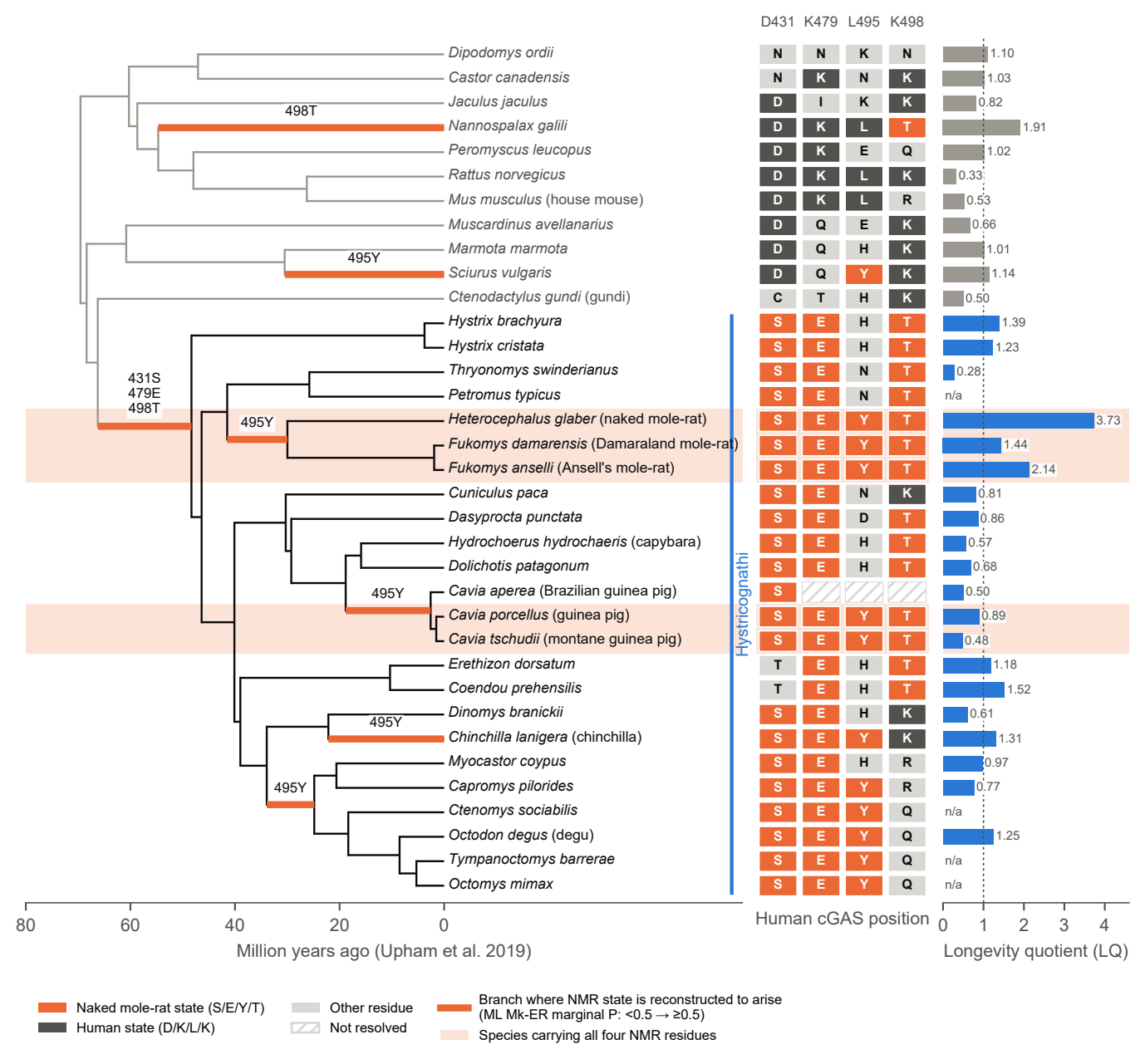

D431 K479 L495 K498
Dipodomys ordii
Castor canadensis
Jaculus jaculus
498T
Nannospalax galili
Peromyscus leucopus
Rattus norvegicus
Mus musculus (house mouse)
Muscardinus avellanarius
Marmota marmota
495Y
Sciurus vulgaris
Ctenodactylus gundi (gundi)
Hystrix brachyura
Hystrix cristata
431S
479E
498T
Thryonomys swinderianus
Petromus typicus
495Y
Heterocephalus glaber (naked mole-rat)
Fukomys damarensis (Damaraland mole-rat)
Fukomys anselli (Ansell's mole-rat)
Cuniculus paca
Dasyprocta punctata
Hydrochoerus hydrochaeris (capybara)
Dolichotis patagonum
495Y
Cavia aperea (Brazilian guinea pig)
Cavia porcellus (guinea pig)
Cavia tschudii (montane guinea pig)
Erethizon dorsatum
Coendou prehensilis
Dinomys branickii
495Y
Chinchilla lanigera (chinchilla)
Myocastor coypus
495Y
Capromys pilorides
Ctenomys sociabilis
Octodon degus (degu)
Tympanoctomys barrerae
Octomys mimax
Hystricognathi
1.10
1.03
0.82
1.91
1.02
0.33
0.53
0.66
1.01
1.14
0.50
1.39
1.23
0.28
n/a
3.73
1.44
2.14
0.81
0.86
0.57
0.68
0.50
0.89
0.48
1.18
1.52
0.61
1.31
0.97
0.77
n/a
1.25
n/a
n/a
80
60
40
20
0
Million years ago (Upham et al. 2019)
Human cGAS position
0
1
2
3
4
Longevity quotient (LQ)
Naked mole-rat state (S/E/Y/T)
Human state (D/K/L/K)
Other residue
Not resolved
Branch where NMR state is reconstructed to arise
(ML Mk-ER marginal P: <0.5 → ≥0.5)
Species carrying all four NMR residues